NARSEE MONJEE INSTITUTE OF MANAGEMENT STUDIES
MPSTME LIBRARY
NMIMS UNIVERSITY

SVKM`S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING
Academic Year: 2023-2024

Program: B Tech/MBA Tech B.TECH INTG Stream: (CSEDS,EXTC,MXTC)/ (DS, MECH,COMP) Year: II/III__ Semester: _III/V

Subject: Management Accounting for Engineers Time: 3 hrs. (10 am to 1 pm)

Date: 21/11/23. No. of Pages: 05

Marks: 100 Marks

Final Examination / Re-Examination (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of the remaining questions, attempt any __4___ questions.
3) In all ___5_ questions to be attempted.
4) All questions carry equal marks.
5) Answer of each new question to be started on a fresh page.
6) Figures in brackets on the right-hand side indicate full marks.
7) Assume Suitable data if necessary.

<table>
<tr><td>Q-1</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>CO- 1; SO- ;
BL- Remember & Understand</td><td>a</td><td>State any FIVE users of financial statements and explain the purposes of their use.</td><td>5</td></tr>
<tr><td>CO-4 ; SO- ;
BL- Evaluate</td><td>b</td><td>A company earned a profit of Rs. 20000 during the year 2022. If the product's variable cost and selling price are Rs.10 and Rs.15 per unit respectively, find out its Contribution, P/V ratio and Margin of Safety.</td><td>5</td></tr>
<tr><td>CO-2 ; SO- ;
BL- Evaluate</td><td>c</td><td>The following particulars about a product are available:
<table><tr><td>Normal usage (per week)</td><td>60 nos.</td></tr><tr><td>Maximum usage (per week)</td><td>80 nos.</td></tr><tr><td>Minimum usage (per week)</td><td>30 nos.</td></tr><tr><td>Reorder quantity</td><td>400 nos.</td></tr><tr><td>Reorder period</td><td>4 to 6 weeks</td></tr></table>Based on the above information, you are required to calculate:
(a) Re-order level
(b) Minimum level
(c) Maximum level
(d) Average level</td><td>5</td></tr>
</table>

CO-4; SO-; BL- Evaluate	d	The standard mix for producing a commodity was as follows: Material A 60 tonnes @ Rs 10 Material B 90 tonnes @ Rs 20 The actual mix used was as follows: Material A 80 tonnes @ Rs 12 Material B 60 tonnes @ Rs 25 Calculate all Material Variances.	5
Q-2 CO-2; SO-; BL- Evaluate	a	The following particulars have been extracted from the books of M/s. Standard Manufacturing Company for the year ended 31-03-2022.	15

Particulars	₹
Opening Stock of Raw Materials	2,35,000
Closing Stock of Raw Material	2,50,000
Raw Material Purchase	10,40,000
Drawing Office Salaries	48,000
Royalty on production	70,000
Carriage Inwards	41,000
Cash Discount Allowed	17,000
Repairs to plant and machinery	53,000
Rent, Rates & Taxes (Factory)	15,000
Rent. Rates & Taxes (Office)	8,000
Office Conveyance	15,500
Salesmen's Salaries & Commission	42,000
Productive Wages	7,00,000
Depreciation on Plant & Machinery	35,500
Depreciation on Office Furniture	3,000
Directors Fees	30,000
Gas and water Charges (Factory)	7,500
Gas and water charges (office)	1,500
Manager's Salaries	60,000
Cost of Catalogues printing	10,000
Loose Tools Written off	8,000
Trade-Fair Expenses	10,000

Out of 48 hours in a week, the Manager devotes 40 hours to the factory and 8 hours to the office per week for the whole year.

The Management has fixed the selling Price @ 110% of the cost.

Prepare a detailed cost sheet for the year ended 31-03-2022.

<table>
<tr><td>Q-2
CO-2; SO-;
BL- Remember</td><td>b</td><td>List down any FIVE points of differences between Activity Based Costing and Traditional Costing.</td><td>5</td></tr>
<tr><td>Q-3
CO-4; SO-;
BL- Evaluate</td><td>a</td><td>A company annually manufactures and sells 20,000 units of a product, the selling price of which is Rs. 50 and the profit earned is Rs. 10 per unit.

The analysis of the cost of 20,000 units is
Material Cost Rs. 3,00,000
Labour Cost Rs. 1,00,000
Overhead (50% variable) Rs. 4,00,000
You are required to compute:
(i) Contribution per unit
(ii) P / V Ratio
(iii) Break Even Sales in Rs.
(iv) Break Even Sales in units
(v) Sales required to earn a profit of Rs. 4,00,000
(vi) Profit when sales are 18,000 units
(vii) Margin of safety when actual sales are Rs. 7,00,000</td><td>15</td></tr>
<tr><td>Q-3
CO-2 SO-; BL-
Understand</td><td>b</td><td>Write short notes on the ABC classification of inventory.</td><td>5</td></tr>
<tr><td>Q-4
CO- 5; SO-;
BL-Apply</td><td>a</td><td>Junoon Ltd. which produces three products, furnishes the following data for the year 2023:
<table>
<tr><th>Particulars</th><th colspan="3">Products</th></tr>
<tr><th></th><th>Alpha</th><th>Beta</th><th>Gamma</th></tr>
<tr><td>Selling Price per unit (₹)</td><td>100</td><td>75</td><td>50</td></tr>
<tr><td>Profit/Volume Ratio</td><td>10%</td><td>20%</td><td>40%</td></tr>
<tr><td>Maximum Sales Potential (units)</td><td>40,000</td><td>25,000</td><td>10,000</td></tr>
<tr><td>Raw Material as % Variable Cost</td><td>50%</td><td>50%</td><td>50%</td></tr>
</table>
The company uses the same raw material for all three products. The raw material is in short supply and the company has a quota for supply of raw material of the value of ₹ 18,00,000 for the year 2023 for manufacture of its products to meet its sales. The total fixed cost is ₹ 6,80,000.
You are required to:
1. Determine a sales mix which will give the maximum overall profit keeping in view the short supply of raw material.
2. Compute the maximum possible profit.</td><td>15</td></tr>
</table>

<table>
<tr><td>Q-4
CO- 4; SO-;
BL-Evaluate</td><td>b</td><td>What is the Break-Even Point (BEP)? Explain the calculation of BEP with the help of an example.</td><td>5</td></tr>
<tr><td>Q-5
CO-3; SO-;
BL-Apply</td><td>a</td><td>A firm expects to have ₹ 25,000 in the bank on 1st May 2022 and requires you to prepare an estimate of cash position during the three months, May-July 2022. The following information is supplied to you:

<table><tr><th>Months</th><th>Sales (₹)</th><th>Purchases (₹)</th><th>Wages (₹)</th><th>Factory (₹)</th><th>Office Expenses (₹)</th><th>Selling Expenses (₹)</th></tr><tr><td>March</td><td>50,000</td><td>30,000</td><td>6,000</td><td>5,000</td><td>4,000</td><td>3,000</td></tr><tr><td>April</td><td>56,000</td><td>32,000</td><td>6,500</td><td>5,500</td><td>4,000</td><td>3,000</td></tr><tr><td>May</td><td>60,000</td><td>35,000</td><td>7,000</td><td>6,000</td><td>4,000</td><td>3,500</td></tr><tr><td>June</td><td>80,000</td><td>40,000</td><td>9,000</td><td>7,500</td><td>4,000</td><td>4,500</td></tr><tr><td>July</td><td>90,000</td><td>40,000</td><td>9,500</td><td>8,000</td><td>4,000</td><td>4,500</td></tr></table>
Other Information:

a) 20% of sales are for cash and the remaining amount is collected in the month following that of sales,
b) Suppliers supply goods at 2 months credit,
c) All expenses are paid in the month following the one in which they are incurred,
d) Income-tax ₹ 25,000 is payable in July,
e) A new plant has been ordered and is expected to be received in June it will cost ₹ 80,000, and
f) The company pays dividends to shareholders and bonuses to workers of ₹ 10,000 and ₹ 15,000 respectively in May.</td><td>15</td></tr>
<tr><td>Q-5
CO-3; SO-;
BL- Understand</td><td>b</td><td>What do you understand by the terms Budget, Budgeting and Budgetary Control? Briefly explain the advantages of budgeting.</td><td>5</td></tr>
<tr><td>Q-6
CO- 3, SO-;
BL- Apply</td><td>a</td><td>The company has prepared the budget for the production of 1,00,000 units of a commodity, for which the budgeted costs are as under:
(Amount in ₹)
<table><tr><td>Raw Material</td><td>2,52,000</td></tr><tr><td>Direct Labour</td><td>75,000</td></tr><tr><td>Direct Expenses</td><td>10,000</td></tr><tr><td>Works Overheads (60% Fixed)</td><td>2,25,000</td></tr><tr><td>Administrative Overheads (80% Fixed)</td><td>40,000</td></tr><tr><td>Selling Overheads (50% Fixed)</td><td>20,000</td></tr></table>
The actual production during the period was only 60,000 units. Prepare a budget showing the total cost per unit for 60,000 units of output.</td><td>12</td></tr>
</table>

Q-6 CO-1; SO-; BL- Remember & Understand	b	Write Short notes on the following: A. Objectives of Management Accounting B. Accounting Concepts – Matching & Going Concern	8
Q-7 CO-4; SO-; BL- Evaluate	a	Calculate Material and Labour Variances from the following data: For 5 units of Product A, the Standard Data are: Material 40 kg. @ Rs 25.00 per kg. Labour 100 hours @ Rs 2.50 per hour Actual data are: Actual Production – 1000 Units. Material 7,840 kg. @ Rs 27.00 per kg. Labour 19,800 hours @ Rs 2.60 per hour	12
Q-7 CO-2; SO-; BL- Evaluate	b	PQR Limited produced a product which has a monthly demand of 52,000 units. The product requires a Components X which is purchased at ₹ 15 per unit. For every finished product, 2 units of Component X are required. The ordering Cost is ₹ 350 per order and the Carrying Cost is 12% p.a. You are required to: 1. Calculate the Economic Order Quality for Component X. 2. Total number of orders to be placed in a year as per EOQ. 3. Time between two orders in the number of days.	8

THE END